

1. Niche dynamics over time

- Fit environmental SDMs
- Identify parsimonious covariates

- Predict on grid
- Calculate experienced environmental conditions as weighted quantiles

- Evaluate conditional effects

- Evaluate trends in experienced environmental conditions over time

2. Drivers of biomass trends in space

- Fit climate-agnostic SDMs with GMRF spatial trends

- Predict on grid
- Extract spatial trends in biomass density

- Fit spatial trends in biomass density to environmental trends and velocities using Random Forest

- Evaluate variable importance, conditional effects, partial dependence plots